

Section: **3.2 – MECHANICALLY FASTENED JOINTS**

Chapter: **3220**

Title: **FASTENERS AND BOLTS FAILURE**

Revision: **1.0**

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1 INTRODUCTION

This chapter presents the failure criterion and expressions to obtain the safety margin of a typical mechanical fastening system: rivet/bolt and nut/collar.

2 METHOD DESCRIPTION

2.1 RIVETS IN TENSION AND SHEAR

From Bruhn's, chapter D1.8, provided the tension and shear allowable loads of the bolt (T_{all} , Q_{all}), as well as the applied ultimate loads (T , Q , calculated by approved methods), are available, the static margin is obtained from the applicable failure indices as follows:

$$(RF \cdot Rt)^2 + (RF \cdot Rs)^3 = 1$$

... where the failure indices are defined as:

$$Rt = T / T_{all}$$

$$Rs = Q / Q_{all}$$

A standard conservative simplification is to modify the exponent on the shear failure index to get a quadratic law with an explicit solution for the Reserve Factor, as:

$$RF = [1 / (Rt^2 + Rs^2)]^{0.5}$$

Nevertheless, for solid rivets (and also for certain types of fasteners) some recognized references recommend the use of special interaction curves for fasteners in combined shear and tension loads. Proposal included in HSB 20210-01 shall be used for those cases.

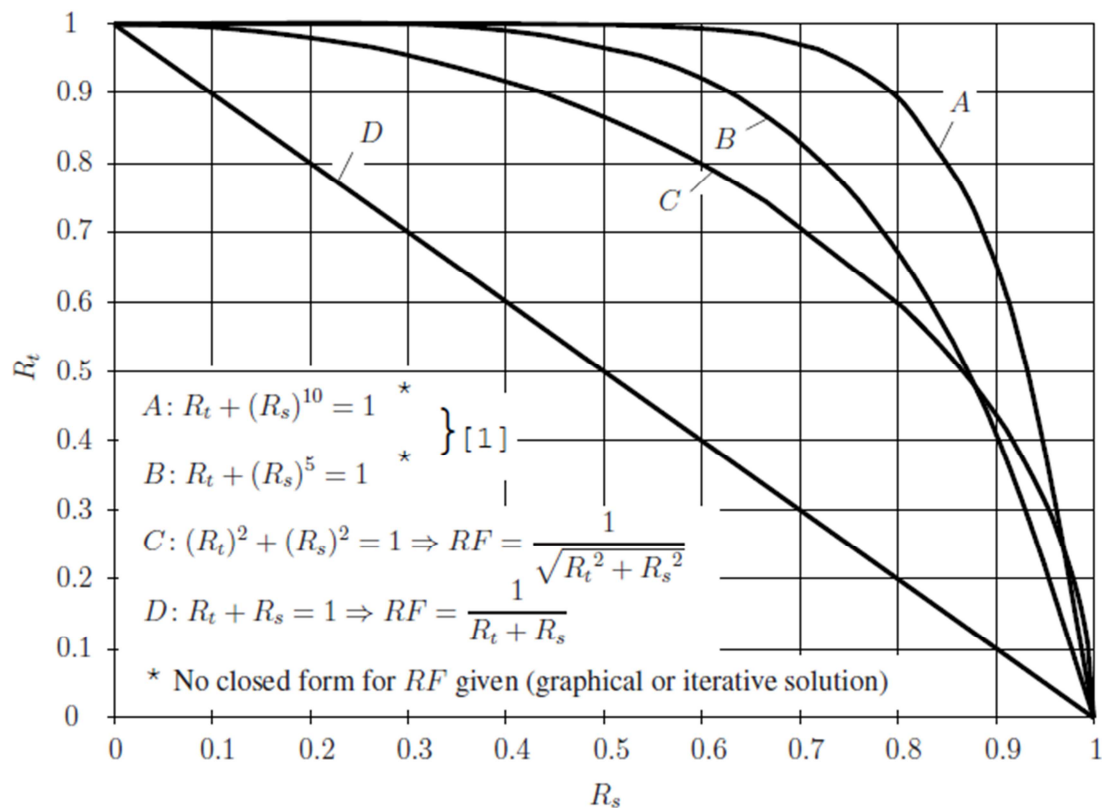


Figure 1. Bolts/rivets interaction curves, extract from HSB 20210-01

Fastener type	Thick components ^a	Thin components ^b	Normal load cases ^c	Crash cases ^d
Hi-Lok, screw (titanium, steel)	Curve A	Curve C	$K = 1.0$	
Hi-Lok (aluminum)	Curve B	Curve C	$K = 1.0$	
swaged collar fastener (e.g. Lockbolt)	Curve D		$K = 0.8$	$K = 1.0$
solid rivet	Curve D		$K = 0.5$	$K = 1.0$
blind rivet / blind bolt	Curve D		$K = 0.2$	$K = 1.0$

^a Shear failure ^b Bearing failure ^c Reduction factor for ultimate tensile load of fastener

Table 1. Curve selection and penalization of fastener tension strength, from HSB 20210-01

Remarks:

- ✓ Note interaction curves A and B are over the curve C within the range where the tension forces are relevant. Thus, using conservatively the quadratic law (curve C), initially suggested in previous page, instead of curves A or B, shall be conservative.
- ✓ For solid rivets, the MMPDS, in table 8.1.2.1(b), proposes some penalization factors to apply to the fastener shear allowable in order to account the potential interaction with the bearing failure for low thickness to diameter ratios. This correction shall not be applied here as the prescribed method to compute bearing strength already accounts for this potential interaction (see chapter §3230 "PLATE FAILURE: BEARING AND PULLTHROUGH").

2.2 NUT

Reserve factor for the nut/collar of a tensioned fastener:

$$RF = T_{all} / T$$

2.3 PRETENSION EFFECT IN HIGH-DUTY TENSION BOLTS

For joints with high-duty tension bolts, the pretension load, and the relationship between applied external force on a fastened joint (normal to attached plates) and the tension in the fastener shank shall have to be considered to calculate the margins.

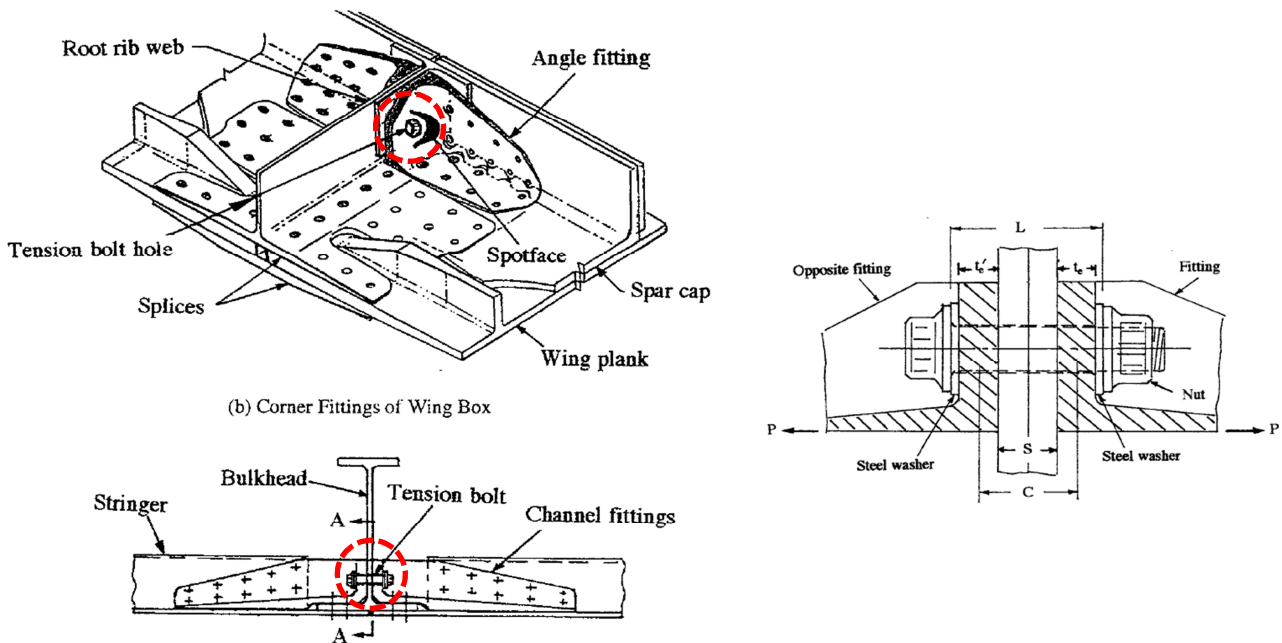


Figure 2. Example of bolted joint in which the effect of the pretension load is relevant (Niu §9.9)

Pretension effect does not need to be considered in riveted/fastened joints transferring mainly shear loads. Examples of fastened joints of this kind can be found in chapter §3210 "TYPES OF RIVETED JOINTS". For cases like those depicted there as *continuous joints* or *plane joints with in-plane loads* there is no need to compute pretension effects in strength analyses.

On the other hand, in bolted attachments of fittings with eccentric loads, the tension and shear forces in one bolt may be both significant at the same time. In such cases, especially if assembly torques are not those prescribed by the standards and/or the tightening torque has to be controlled along service life for any reasons, computing the effect of pretension in the calculation of the real load seen by the bolt, and in the expressions for obtaining the reserve factors, will be necessary. Examples of attachments of fittings with eccentric loads can be found also in chapter §3210.

2.3.1 PRETENSION AND EFFECT IN BOLT TENSION

The relationship between tightening torque applied on a threaded fastener and the resulting pretension load on it, as well as the method to calculate the bolt tension from external load accounting the installation pretension are explained (among other references) in HSB 24000-02 (Ref. 3). This methodology shall be the basis of the Airbus DS methodology to account this effect.

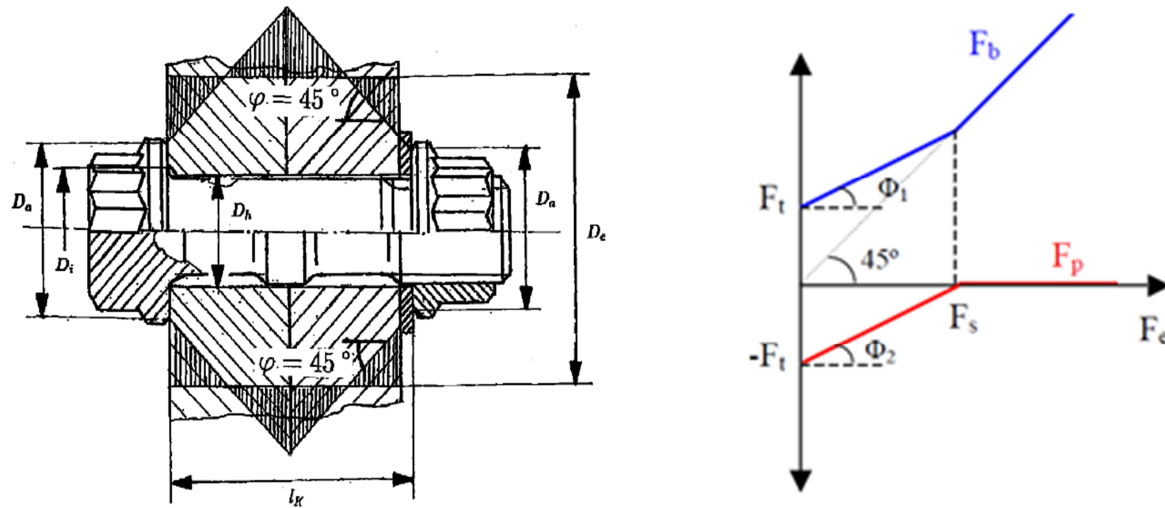


Figure 3. Plates stiffness approach (HSB 24000-02) & typical curve external load vs bolt tension

Relationships between bolted joint materials, geometry and external applied loads with bolt tension, as well as with the rest of magnitudes related to pretension effects (including the plates separation force and the slope of the curve relating applied external load and bolt tension), shall be obtained using the above mentioned HSB 24000-02 method.

Symbol in HSB24000-02	Symbol in graph & formulas	Description
F_A	F_e	Applied external load in bolt axis direction
F_M	F_t	Pretension load
F_S	F_b or T	Bolt tension load (total)
F_{SA}		Difference between bolt tension and pretension $F_S = F_M + F_{SA} = F_M + \Phi_n \cdot F_A \leftrightarrow F_b = F_t + \tan \Phi_1 \cdot F_e$
Φ_n	$\tan \Phi_1$	Slope of the curve relating bolt-tension vs applied-load, within the range before plates separation
	F_s	Tension force required for plates separation. If using HSB notation that is the value of F_A when the clamp force is zero... $F_{KR} = F_M - F_{PA} = 0 = F_M - (F_A - F_{SA})$

Table 2. Different notations in HSB and this document in discussion about pretension analysis

2.3.2 CALCULATION OF RESERVE FACTORS

As highlighted by previous figure, in the region starting from zero load (pretension in bolt) up to the force that separates the plates, the tension in the bolt is not directly proportional to the “external” applied force. This non-linear (bilinear indeed) relationship derives in two different formulations for calculating the Reserve Factors, which must be referred to the “external” forces, depending on the region in which failure occurs.

The classical formulation for relating the tension force in the bolt with the applied force normal to plate (notation in line with previous figure, slightly different than that used in HSB 24000-02) is as follows:

$$\begin{aligned} F_b &= F_t + (\tan \Phi_1) \cdot F_e & \text{if } F_e < F_s \text{ (before plates separation)} \\ F_b &= F_e & \text{if } F_e > F_s \text{ (after plates separation)} \end{aligned}$$

It is remarked that, from this point, the presented discussion and formulas exceed the scope of the HSB 24000-02 method description.

In order to cover also cases with shear in the bolt, the quadratic failure criterion proposed for typical threaded bolts is assumed for simplification (applicable to threaded bolts and Hi-Lok/Hi-Lite fasteners, see discussion in sub-chapter §2.1):

$$1 = R_t^2 + R_s^2$$

Finally, the pretension force is considered a fixed parameter in this problem, treated as geometry or material properties. Then, at failure load levels, shear and tension forces in bolt can be expressed as:

$$\begin{aligned} Q_u &= RF \cdot Q \\ F_{bu} &= F_t + RF \cdot (\tan \Phi_1) \cdot F_e & \text{if } F_e < F_s \text{ (before plates separation)} \\ F_{bu} &= RF \cdot F_e & \text{if } F_e > F_s \text{ (after plates separation)} \end{aligned}$$

Bolt/nut failure BEFORE plates separation

As in this range the tension in the bolt T is not directly proportional to the “external” force F_e , the calculation of the Reserve Factor, which must be referred to those “external” forces, becomes not as straightforward...

- Nut:

$$F_{bu} / T_{all-n} = (F_t + RF_n \cdot F_e \cdot \tan \Phi_1) / T_{all-n} = 1 \quad \rightarrow \quad RF_n = (T_{all-n} - F_t) / (F_e \cdot \tan \Phi_1)$$

- Bolt:

$$(Q_u / Q_{all-b})^2 + (F_{bu} / T_{all-b})^2 = [RF_b \cdot Q / Q_{all-b}]^2 + [(F_t + RF_b \cdot F_e \cdot \tan \Phi_1) / T_{all-b}]^2 = 1$$

$$\rightarrow RF_b = [-B + (B^2 - A \cdot C)^{0.5}] / A$$

...where:

$$A = (Q / Q_{all-b})^2 + (F_e \cdot \tan \Phi_1 / T_{all-b})^2$$

$$B = F_t \cdot F_e \cdot \tan \Phi_1 / T_{all-b}^2$$

$$C = (F_t / T_{all-b})^2 - 1$$

Bolt/nut failure AFTER plates separation

This is the most simple case, as beyond plates separation the tension in the bolt T is equal to the “external” applied force normal to the plates F_e , thus:

- Nut: $RF_n = T_{all-n} / F_e$
- Bolt: $RF_b = 1 / [(Q / Q_{all-b})^2 + (F_e / T_{all-b})^2]^{0.5}$

Failure load and Reserve Factor

As failure instant (before or after plates separation) is initially unknown, dummy failure loads shall be calculated using both formulations, i.e.

$$F_1 = RF_1 \cdot F_e \quad \text{with } RF_1 \text{ calculated using the formulation for load BEFORE plates separation}$$

$$F_2 = RF_2 \cdot F_e \quad \text{with } RF_2 \text{ calculated using the formulation for load AFTER plates separation}$$

Those values of dummy failure loads not consistent with the formulation used (e.g. F_1 greater than separation force or F_2 lower than separation force) are then discarded as not possible. The remaining failure load and RF must be the right values. If both dummy failure loads were consistent with the formulation used, the lowest (first failure) must be the right one.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
b / bolt	-	As subscript, referred to bolt
F_b / T	N	Tension force load in bolt
F_e	N	External applied force normal to attached plates
F_s	N	Separation force (first load level at which contact force between plates become 0)
F_t	N	Bolt pretension
F_p	N	Contact/compression force in plates at pretensed joints
n / nut	-	As subscript, referred to nut
Q	N	Shear load
Q_{all}	N	Shear allowable load
RF	-	Reserve Factor, i.e. ratio between allowable load and applied load
R_s	-	Ratio between applied shear and shear allowable load
R_t	-	Ratio between applied tension and tensile allowable load
T_{all}	N	Tension allowable load
u	-	As subscript, indicating failure (ultimate) load level
Φ_1	°	Angle of the slope of the function relating the applied external load with bolt tension in the range before plates separation

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 2	[SAE] MMPDS	9	Metallic Materials Properties Development and Standardization
Ref. 3	HSB. 24000-02	E	Analysis of high duty bolted joints with one concentric cylindrical bolt
Ref. 4	[M.C.Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 5	HSB 20210-01	Is. D	Reserve factors of fasteners under combined shear and tension loading

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures (Modules HSB26101, HSB24000/PRETENSION & BOLTS3D)

Table 3. *Software programs referenced in this chapter*

DISTRIBUTION LIST

<i>Department</i>	<i>Name</i>	<i>Full Doc.</i>	<i>Cover Page</i>
Stress Methods & Optimisation (TEASP-TL3)	Dr. Fernað Daoud		X
Stress Methods & Optimisation (TEASP-TL3)	Dr. Alexander Hörmann	X	
Stress M&L, Derivatives, Mission (TEASA-TL2)	Miguel Angel Calero Gallardo	X	
Stress Combat MAN (TEASA-TL3)	Sebastian Rönz	X	
Stress Combat, Aerostructures, A400M and Technologies GET (TEASA-TL4)	Sofia Ponce Borrero	X	
Structural Repair GET (TEASA-TL6)	Jose Luis Melchor Martin de la Sierra	X	

Table 4. *Distribution list (of electronic document, ahead of public release)*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 26/06/2019	G. Baños	First issue	All pages

Table 5. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Analysis (TEASA)	Miguel Ángel Calero 26/06/2019	Bernhard Mayer 26/06/2019
Stress Methods (TEASP-TL3)	G. Baños 26/06/2019	Florian Glock 26/06/2019

Table 6. *Approval signatures table for last revision of this chapter*